

Unit 5: Network Forensics

M.Sc. DFIS | Network Security | NFSU Delhi Campus

School of Cyber Security and Digital Forensics

Unit Overview

What this unit covers

*Digital Evidence · Investigation Methodology · Evidence Acquisition
Traffic Capture · Log Analysis · Data in Motion*

Unit 5 Learning Objectives

By the end of this unit, students will be able to:

- Define and classify types of digital and network-based evidence
- Apply the OSCAR investigative methodology to network incidents
- Conduct passive and active evidence acquisition from network sources
- Use tcpdump and Wireshark for traffic capture and analysis
- Interpret event logs and correlate evidence across sources

Tools Covered in This Unit

- tcpdump
- Wireshark
- libpcap / WinPcap
- Berkeley Packet Filter (BPF)
- Snort (IDS / packet logger)
- SIEM platforms (Splunk, QRadar, Sentinel)

What is Digital Evidence?

Digital evidence is any information stored or transmitted in binary form that may be relied upon in court.

Key Characteristics

- **Admissibility**
- **Authenticity**
- **Integrity**
- **Reliability**

"Evidence is only as good as its chain of custody"

Network-based digital evidence is a subset of digital evidence — produced as a result of communications over a network and governed by the same admissibility standards.

Types of Evidence – Overview

Type	Definition	Example
Real	Physical, tangible item	Hard drive, USB, fingerprint
Best	Most authoritative version available	Original file, bit-for-bit image
Direct	Directly asserts the fact	Eyewitness testimony
Circumstantial	Allows inference, does not directly prove	Authorship of emails, chat logs
Hearsay	Second-hand testimony	Forwarded email chains
Business Records	Routinely generated, sufficiently reliable	Access logs, authentication logs
Digital	Stored or transmitted in binary form	Files, logs, images, metadata
Network-Based Digital	Produced as result of network communications	Packet logs, DHCP logs, IM sessions

Real Evidence and Best Evidence

Real Evidence

- Physical, tangible items presented in court
- Directly perceived by judges or juries without interpretation

Examples:

- Hard drive or USB device
- Signed contract or printed document
- Fingerprint card
- Server rack physically seized from a crime scene

Best Evidence

- The most authoritative version of evidence available
- Courts prefer originals; copies require justification

Examples:

- Original digital file (not a scan of a printout)
- Bit-for-bit forensic snapshot of a drive
- Original photograph of a crime scene

Direct and Circumstantial Evidence

Direct Evidence

- Directly asserts the fact to be proved
- Generally the most persuasive category
- No inference required to reach the conclusion

Example:

- *"I personally observed the accused using a password-cracking tool on the server."*

Circumstantial Evidence

- Does not directly prove a fact — allows inference
- Can still be decisive when accumulated

Frequently used to:

- Establish authorship of emails or chat messages
- Link a suspect to digital artefacts
- Demonstrate access or opportunity

Hearsay and Business Records

Hearsay Evidence

- Second-hand testimony — not from a direct witness
- Generally inadmissible unless an exception applies

Digital hearsay examples:

- Text files containing personal statements
- Forwarded email chains
- Screenshots of conversations

Business Records

- Routinely generated and retained documents
- Accepted as sufficiently reliable because of regularity

Common examples:

- Server access logs
- Authentication and login records
- Invoices and financial records
- */var/log/messages on Linux systems*

Network-Based Digital Evidence

"The network is a witness that never lies — but it must be captured before it disappears"

Definition:

Evidence produced as a result of communications over a network — including emails, IM sessions, browser activity, web-based email, packet logs, DHCP logs, and authentication logs.

Common Examples

- Email headers and message content
- Instant messaging session logs
- Full packet capture (PCAP) files
- DHCP lease assignment records

Key Challenge

Volatile by nature — network data exists only while it is in transit. It must be acquired in real time or it is lost permanently. Unlike files on disk, it cannot be recovered after the fact.

Challenges in Network Evidence Acquisition

Acquisition

Locating specific evidence in distributed environments is difficult. Evidence may span multiple network segments, jurisdictions, and device types.

Content

Metadata may be absent; network devices have limited storage capacity and rarely retain full payload data for extended periods.

Storage

Most network devices lack persistent secondary storage. Routers and switches do not maintain detailed event logs by default.

Privacy

Jurisdictional and legal constraints restrict full packet capture. Investigator must operate within applicable legal authority.

Seizure

Taking a network device offline to seize it can disrupt entire network segments — potentially harming business operations.

Admissibility

Legal precedents for network-based evidence are still evolving in many jurisdictions including India.

The Investigative Framework

OSCAR Methodology

Obtain · Strategize · Collect · Analyze · Report

OSCAR: The Investigative Methodology

O	S	C	A	R
Obtain	Strategize	Collect	Analyze	Report
Gather all facts about the incident and environment	Plan the investigation approach and prioritise sources	Acquire, document, and preserve evidence lawfully	Correlate, timeline, and interpret collected evidence	Produce a clear, defensible, factual report

O: Obtain Information

The Incident

- Description and nature of the incident
- Date and time of discovery
- Persons involved (victim, suspect, witness)
- Systems and data affected
- Actions taken since discovery
- Incident manager identity and authority
- Legal and regulatory issues
- Investigation time frame and goals

The Environment

- Business model and organisational context
- Legal and jurisdictional constraints
- Network topology and architecture
- Available evidence sources
- Organisational structure (who reports to whom)
- Communications systems in use
- Available investigative resources

S: Strategize — Building the Investigative Plan

Building the Investigative Plan

- Understand the goals and time frame of the investigation
- List all available resources: personnel, equipment, time, budget
- Identify all probable evidence sources across the network
- Estimate value and cost of acquiring each evidence source
- Prioritise evidence acquisition — highest value, least disruption first
- Plan the initial acquisition and analysis sequence
- Set a communication and update schedule for stakeholders
- Document the plan formally before beginning

Remember: forensic investigation is iterative. New evidence will change priorities. Expect to revisit sources and revise the plan as the investigation develops.

C: Collect Evidence

1

Document

Keep a detailed log of every system accessed, every action taken, every tool used, and every person involved. Contemporaneous notes are essential.

2

Capture

Capture packets to hard drive. Copy logs to HDD or CD. Image web proxies and logging servers. Photograph physical evidence.

3

Store and Transport

Use write-protected, secure storage. Maintain an unbroken chain of custody. Every access must be signed and verifiable.

Best Practices: Acquire as soon as possible and lawfully. Make cryptographically verifiable copies (MD5 + SHA-256). Sequester originals. Analyse only copies. Use reputable, documented tools. Record every command.

A: Analyze

Analysis Steps

Correlation

- Link data across timestamps and sources

Timeline Construction

- Establish full chronology of events

Events of Interest

- Isolate high-significance events from background noise

Corroboration

- Verify findings, rule out false positives

Recovery

- Look for additional evidence suggested by initial findings

Interpretation

- Draw defensible, evidence-based conclusions

"Analysis is only as strong as the evidence collection that precedes it."

The analysis phase is not independent — it depends on the completeness and integrity of the Collect phase. Gaps in collection produce gaps in analysis that can never be recovered.

R: Report

Report Requirements

- Understandable by non-technical readers
 - Legal teams, managers, HR, judges, juries
- Defensible in every detail
 - Every conclusion must link to evidence
- Factual — not speculative
 - Do not draw conclusions beyond what the evidence supports

Audience-Specific Needs

Legal team:

Admissibility, chain of custody, lawful acquisition

Management:

Business impact, risk exposure, remediation steps

Technical review:

Methodology, tool outputs, reproducibility

Courts / tribunals:

Neutrality, reproducibility, expert qualification

Evidence Acquisition

Sources and Methods

Physical Interception · RF Capture · Software-Based Acquisition · Active vs Passive

Passive vs Active Acquisition

Passive Acquisition

- Gather evidence without emitting data at Layer 2 and above
- No modification to network traffic
- Preferred for court admissibility

Methods:

- Port mirroring (SPAN port)
- Optical tapping (fibre)
- Promiscuous mode NIC capture

Active Acquisition

- Collecting evidence by interacting with network devices
- Inherently modifies the environment
- Sometimes unavoidable (live systems, routers)
- Must be carefully documented to maintain validity

Examples:

- Querying a live router for routing table
- Issuing commands to running servers

"Every investigator interaction with a live system modifies it — document the footprint."

Physical Interception: Copper Cabling

Coaxial Cable

- Historically used in bus topology (10BASE5, 10BASE2)
- Shared medium — all stations receive all frames
- Tapping the copper core provides access to all station traffic
- Vampire tap clamps onto cable without cutting
- Rarely seen in modern networks but present in legacy/industrial environments

Twisted Pair (TP)

- Standard medium in modern switched Ethernet
- Commercial inline tap captures voltages on all twisted pairs
- Tap is inserted by physically separating the cable at a point

Key point:

- **Inline tap insertion causes a brief network disruption when the cable is separated**

Inline Network TAP

Definition and Configuration

- Hardware device inserted between two network devices (e.g. switch and router)
- Copies all traffic without compromising normal network operation
- Typical configuration: four ports
 - Two inline ports — carry normal network traffic
 - Two sniffing ports — mirror traffic (one per direction)
 - The capture device connects to the sniffing ports only

Practical Concern

Every additional cable break is a potential point of failure. TAP insertion inherently increases physical risk and must be documented.

Forensic Implication

TAP insertion requires a brief cable separation — document the exact time, duration, and individuals present at the point of insertion.

Vampire Taps

Definition

- Connectors that clamp onto and physically bite into 10BASE5 (thick Ethernet) coaxial cabling
- Allow a station to connect without cutting the cable
- Named for the way they 'pierce' the cable jacket and core
- Used extensively in early corporate and government networks

Forensic Relevance

- Rarely encountered in modern networks
- May appear in legacy, industrial, or government network investigations
- Leaves physical evidence of intrusion on the cable jacket
- Useful indicator of physical tampering in legacy infrastructure audits

Fibre Optic Interception

Fibre Bending Attack

- Attacker slightly bends the optical fibre at a point
- Light leaks at the bend due to the physics of total internal reflection
- A sensor positioned at the bend captures the leaked signal
- Minimal signal loss if performed carefully
- Difficult to detect without an OTDR baseline measurement

Inline Optical TAP

- Technicians splice the optical cable and connect to each tap port
- Causes detectable network disruption at time of insertion
- May cause noticeable signal degradation if not properly aligned

Detection methods:

- Optical Time-Domain Reflectometer (OTDR)
- Measures signal quality and detects anomalies along the fibre

Radio Frequency (WLAN) Evidence Acquisition

Why WLAN is Different

RF is a shared medium by nature — physical segmentation is impossible. Passive acquisition is very easy for both investigators and adversaries. No physical connection needed.

Evidence Available from 802.11 Management Traffic

Even when data is encrypted: broadcast SSIDs, WAP MAC addresses, supported encryption and authentication algorithms, associated client MAC addresses, and in many configurations full Layer 3+ packet content.

Investigator Requirements

Compatible 802.11 adapter (a/b/g/n/ac as applicable). Monitor mode capability. Knowledge of applicable encryption: WEP (crackable with sufficient traffic), WPA/WPA2-PSK (capture 4-way handshake and perform offline dictionary attack).

Software Tools for Network Forensics

Traffic Capture and Analysis

libpcap · WinPcap · tcpdump · Wireshark · Snort · BPF

libpcap and WinPcap

libpcap (UNIX)

- UNIX C library providing an API for capturing and filtering Layer 2 frames
- Provides portability across different UNIX architectures
- Captures at Layer 2 from any suitable network interface
- Stores captured data in PCAP format

Foundation for:

- tcpdump, Wireshark, Snort, nmap, ngrep, Zeek

WinPcap / Npcap (Windows)

- Published 1999 by NetGroup, Politecnico di Torino
- Windows port of libpcap
- Same PCAP format compatibility
- WinPcap largely superseded by Npcap
- Npcap is maintained by the Nmap Project
- Npcap adds NDIS 6 support and better Windows 10/11 compatibility

"The PCAP file format is the universal lingua franca of network forensics — understood by virtually every analysis tool."

tcpdump: Introduction and Purpose

Command-line packet capture tool built on libpcap

- Captures traffic bit-by-bit as it traverses the physical media
- Decodes Layer 2 through Layer 4 protocols (Ethernet, IP, TCP, UDP, ICMP)
- Output formats: hexadecimal, ASCII, or both
- Can write raw capture to PCAP files for later analysis in Wireshark
- Forensic strength: capable of capturing with sufficient fidelity for court-admissible evidence
- WinDump: direct port of tcpdump for Windows, same syntax

Forensic Strength

Capable of capturing with the fidelity required for court-admissible evidence when used correctly.

Key Limitation

On high-traffic networks, capture may be limited by processor speed and available disk space.

tcpdump: Snapshot Length (snaplen)

snaplen

The number of bytes of each captured frame that tcpdump actually records to disk

Too short: Data permanently lost from every captured frame — upper-layer content is truncated

Too long: Performance degradation; limits the volume of storable traffic

Historical note: Ethernet MTU is 1500 bytes; frame with header is 1514 bytes — once the recommended snaplen

Modern practice: Use snaplen = 0 to automatically capture the entire frame regardless of length (tcpdump -s 0)

In forensic captures, always use -s 0 (snaplen = 0). Truncated evidence is incomplete evidence and may be challenged in court.

tcpdump: Command Reference

Flag	Purpose
-i	Listen on specified interface (eth0, en1, wlan0)
-n	Do not resolve addresses to names (faster, no DNS queries)
-r	Read packets from a saved PCAP file
-w	Write captured packets to a PCAP file
-s	Set snapshot length (use -s 0 for full frame capture)
-C	Limit capture file size; start a new file when limit exceeded
-W	Combined with -C: limit number of files; rotate when limit reached
-D	List available adapters (WinDump on Windows only)

Example: `tcpdump -i eth0 -s 0 -w capture.pcap`

tcpdump: Practical Filtering Examples

Filtering reduces noise — focus acquisition on what matters

Write all traffic on eth0 to file:

```
tcpdump -i eth0 -w full_capture.pcap
```

Capture all traffic except HTTP:

```
tcpdump -nni eth0 'not (tcp and port 80)'
```

Full capture of traffic to/from a specific host:

```
tcpdump -i eth0 -s 0 -w host.pcap 'host 10.10.10.10'
```

Capture packets with IP options set (IHL > 20 bytes):

```
tcpdump -i eth0 'ip[0] & 0x0F > 0x05'
```

Always record in the evidence log: exact command used, interface name, capture start time, capture end time, and operator name.

Berkeley Packet Filter (BPF): Introduction

What is BPF?

- Extremely powerful filtering language included with libpcap
- Filters traffic based on field values at Layers 2, 3, and 4
- Can range from simple primitives to complex bitmask operations
- Filters can be chained with logical AND and OR operators
- Applied at capture time — filtered frames are never stored
- Same BPF syntax works across tcpdump, Wireshark capture filters, and Snort

Three Qualifier Types

Type qualifiers:

host, net, port, portrange

Direction qualifiers:

src, dst, src or dst, src and dst

Protocol qualifiers:

ether, ip, ip6, arp, tcp, udp, wlan

BPF filters are applied at the kernel level before packets reach user space — making them extremely efficient even on high-speed links.

BPF Primitives Reference

Primitive	Usage
host id	Match source or destination IP address
src host id / dst host id	Match traffic in a specific direction
net id	Match an entire subnet (e.g. net 192.168.1.0/24)
port id / src port / dst port	Filter by port number
ether host id	Match by MAC (hardware) address
tcp / udp / icmp / arp	Filter by protocol type
vlan id	Filter by VLAN tag identifier
gateway id	Match packets routed through a specific gateway

Example BPF

expression:

```
'host 192.168.0.1 and not host 10.1.1.1 and (port 138 or port 139 or port 445)'
```

BPF: Filtering by Byte and Bit Value

Byte Value Filtering

- Syntax: protocol[offset] or protocol[offset:length] — Byte offsets start at 0
- **ip[9] != 1** — Exclude ICMP (protocol number field = 1)
- **tcp[0:2] = 31337** — Match source port 31337
- **ip[12:4] = 0xC0A80101** — Match source IP 192.168.1.1

Bitmask Filtering

- Use the & operator to isolate specific bits within a byte
- **ip[0] & 0x0F > 0x05** — Match packets where IP header is greater than 20 bytes (IP options present)
- *Explanation: the low-order nibble of IP byte 0 is the IHL field. IHL of 5 means 5 words x 4 = 20 bytes (standard). IHL > 5 means options are present.*

Wireshark: Introduction

What Wireshark Is

- GUI-based packet analyser built on libpcap
- The most widely used network protocol analyser in the world
- Supports deep decode of hundreds of protocols
- Captures live traffic or reads existing PCAP files
- Can follow TCP streams to reconstruct sessions
- Has a powerful display filter language (distinct from BPF)

Key Forensic Capabilities

- Full packet decode to application layer
- TCP stream reconstruction (HTTP, FTP, email)
- File extraction from captured sessions
- Statistics, IO graphs, and endpoint summaries
- Export filtered packets to a new PCAP file
- Colouring rules to highlight traffic of interest

Wireshark: Display Filters vs Capture Filters

Capture Filters (BPF)

- Applied at the time of capture
- Reduces volume of stored data
- Uses BPF syntax (same as tcpdump)
- **WARNING: Filtered-out packets are lost forever — they are never stored**
- Use only when storage constraints force it

Display Filters

- Applied after capture, to already-stored PCAP data
- Does NOT discard packets — only hides them from view
- Uses Wireshark's own display filter syntax

Examples:

- `http`
- `tcp.port == 443`
- `ip.addr == 10.0.0.1`

"For forensic work, capture everything and filter for display — never pre-filter unless storage constraints force it."

Wireshark: Common Display Filters

Filter Expression	What It Shows
<code>http</code>	All HTTP traffic
<code>tcp.port == 443</code>	HTTPS traffic
<code>ip.addr == 10.1.1.5</code>	All traffic to or from that IP address
<code>dns</code>	All DNS queries and responses
<code>tcp.flags.syn == 1</code>	TCP SYN packets (connection initiation attempts)
<code>ftp</code>	FTP control session commands
<code>smtp</code>	Email transmission (SMTP protocol)
<code>tcp.analysis.retransmission</code>	Retransmitted packets (indicates network issues)
<code>!(arp or dns or icmp)</code>	Exclude routine background noise traffic

Logs as Forensic Evidence

Event Log Analysis

Sources · Aggregation · Correlation · Timeline Reconstruction

Sources of Network-Based Log Evidence

Infrastructure Logs

- Firewall logs (allow/deny events, source/destination IP and port)
- Router and switch logs (interface events, routing changes)
- DHCP lease assignment logs (MAC to IP mapping with timestamps)
- DNS query logs (which host resolved which domain and when)

Authentication and Access Logs

- Active Directory / LDAP authentication events
- VPN concentrator connection logs
- SSH access logs (/var/log/auth.log, /var/log/secure)
- Web proxy access logs (URL, user agent, response code)

Application and Platform Logs

- Email server logs (SMTP relay, sender, recipient, message ID)
- Web server access logs (Apache, Nginx, IIS)
- IDS alerts (Snort, Suricata)
- Wireless AP association and authentication logs

Event Log Aggregation: SIEM

What is a SIEM?

- Security Information and Event Management
- Centralises log collection from all sources into one platform
- Normalises logs from disparate formats into a common schema
- Enables real-time alerting and long-term log retention

Examples:

- Splunk, IBM QRadar, Microsoft Sentinel, Elastic SIEM

Why Aggregation Matters

- Single pane of glass across all evidence sources
- Timestamps normalised to UTC for accurate correlation
- Logs survive even if the source device is wiped or offline
- Enables correlation across sources that individually reveal nothing
- Dramatically reduces investigator time versus manual log review

Event Log Correlation

1

Normalise

Convert all log timestamps to UTC.
Standardise field names across different log sources and formats.

2

Link

Join records by a common field: IP address, MAC address, username, session ID, or request ID.

3

Sequence

Arrange events in chronological order to build the complete attack timeline from first contact to last action.

4

Interpret

Identify gaps, anomalies, and patterns.
Distinguish attacker activity from background noise and legitimate user activity.

"Correlation is the forensic equivalent of assembling a jigsaw puzzle — each log source is one piece."

Data in Motion Investigation

What is Data in Motion?

- Data actively traversing a network — as opposed to data at rest on storage media

Examples:

- Live packet streams
- Real-time VoIP sessions
- File transfers in progress
- Command-and-control traffic
- **Forensically volatile — if not captured in real time, it is permanently lost**

Investigative Approaches

- Port mirroring (SPAN port) on managed switches
- Inline hardware TAP devices
- Full packet capture appliances (SecurityOnion, ExtraHop)
- Selective capture using BPF filters
- NetFlow and IPFIX for metadata without full payload capture

NetFlow and IPFIX: Traffic Metadata

What NetFlow Records (per flow)

- Source and destination IP addresses
- Source and destination port numbers
- Protocol type (TCP, UDP, ICMP)
- Total packets and bytes transferred in the flow
- Flow start and end timestamps
- Ingress interface identifier

Forensic Value of NetFlow / IPFIX

- Does NOT capture payload — only metadata (privacy-preserving)
- Extremely lightweight — can be retained for months on modest storage
- Reveals who talked to whom, when, for how long, and how much data was transferred
- Sufficient to establish communication patterns even without payload access
- Valuable when full packet capture is not feasible

Applying the Methodology

Case Studies in Network Forensics

Three real-world network forensic investigations

Case Study 1: Hospital Laptop Theft

Scenario	A doctor's laptop stolen from a metropolitan hospital. Hard drive unencrypted; contains patient PHI including lab results, schedules, and diagnoses. Hospital subject to HIPAA and HITECH obligations.
Challenge	Determine precise time of theft, track device, identify data at risk, verify whether stolen credentials were later used for network access.
Evidence Sources	Wireless access point logs, DHCP logs, Active Directory authentication logs, VPN connection logs, email server logs, physical surveillance footage.
Finding	WAP logs pinpointed theft time and tracked laptop movement to parking garage. No post-theft network access detected. Parking camera provided image used by police to identify suspect.
Lesson	WAP logs are underutilised evidence sources. Log retention policy directly determined what was recoverable — short retention = less evidence.

Case Study 2: Corporate Piracy Detection

Scenario	A global corporation's centralised IDS alerts on peer-to-peer file sharing. Filenames reference movies still in theatrical release.
Challenge	Identify physical location of P2P traffic source, identify the specific user, and determine exactly what was shared.
Evidence Sources	IDS alerts, DHCP lease logs, switch port mapping, full packet capture from IDS sensor, hard drive of suspect desktop.
Finding	IP address traced via DHCP to MAC address, MAC to switch port, switch port to physical desk location. Packet capture allowed analysts to reconstruct and play back infringing video files. Hard drive corroborated with matching files.
Lesson	DHCP logs are the critical link between IP addresses and physical machines. Packet reconstruction can produce court-ready evidence.

Case Study 3: Hacked Government Server

Scenario	Routine AV scan on a state government server detects rootkit files in a local administrator home directory. Server is on a subnet containing Social Security numbers and financial data.
Challenge	Confirm compromise, determine how it occurred, check whether other servers were affected, determine if sensitive data was exfiltrated.
Evidence Sources	Antivirus alerts, SSH logs from central logging server, firewall configuration, network documentation, centralised authentication logs, server hard drive.
Finding	SSH brute-force attack exploited a weak password on an old administrator account. Firewall misconfiguration allowed external SSH in violation of policy. Attacker access was brief; no data was exfiltrated; other servers unaffected.
Lesson	Centralised logging made the investigation possible. Firewall configurations must be audited regularly. Old accounts must be purged promptly.

Comparing the Three Cases: Key Takeaways

Case	Primary Evidence Source	Key Tool / Method	Core Lesson
Hospital Laptop	Wireless AP logs	Log correlation across WAP, DHCP, AD	Retention policy determines recoverability
Corporate Piracy	DHCP logs + packet capture	tcpdump / IDS + file carving	Packet reconstruction produces court-ready evidence
Government	SSH logs + firewall logs	Centralised SIEM / Log server	Logging infrastructure is the investigator's

Across all three cases, the key variable was not the sophistication of the attacker — it was the quality and retention of logs. Logging infrastructure is an investment in future investigability.

Chain of Custody in Network Forensics

Why It Matters

- Network evidence is easily challenged in court if integrity cannot be demonstrated
- Chain of custody documents every person who has touched the evidence
- And every action taken on or with it
- A break in the chain creates grounds for challenging admissibility

Practical Steps

- Hash the captured PCAP immediately (MD5 and SHA-256)
- Record hash value, filename, file size, interface, tool version, operator name
- Store original in write-protected media or evidence vault
- Work only on verified copies — never the original
- Rehash copies and verify against original before each analysis session
- Log every tool used and every command executed

Legal and Ethical Considerations

Authorisation

- Always obtain written authorisation before capturing any traffic
- Corporate environments: authorisation from appropriate management level
- Law enforcement: warrant or lawful intercept authority required
- Unauthorised interception can render evidence inadmissible and expose the investigator to criminal liability

Privacy

- Network capture captures everyone's traffic — not just the suspect's
- Minimisation obligations: capture only what is necessary to the investigation
- Jurisdictional variation: rules differ significantly across India, US, EU, and other jurisdictions

Admissibility Requirements

- Evidence must be authentic, reliable, and complete
- Tool outputs must be reproducible and independently verifiable
- Investigator must be qualified to testify on methods used

Practical Lab: Traffic Analysis with tcpdump and Wireshark

Exercise Objective

Students capture live traffic in a controlled lab environment, apply BPF filters, and use Wireshark to reconstruct an HTTP session and extract a file from the capture.

Lab Tasks (Step by Step)

1. Start tcpdump on eth0 with snaplen 0, writing to a PCAP file
2. Open a browser and visit an HTTP (non-HTTPS) test site
3. Stop capture. Open PCAP in Wireshark.
4. Apply display filter: http
5. Follow the TCP stream of a GET request
6. Use File > Export Objects > HTTP to extract any transferred file
7. Hash the extracted file (MD5 + SHA-256) and document
8. Apply BPF filter to re-read PCAP for only traffic to the target IP

"If you cannot explain every step of what your tool did, you cannot defend your evidence in court."

Snort: Brief Introduction

What Snort Is

- Open-source network intrusion detection and prevention system (IDS/IPS)
- Built on libpcap — same capture layer as tcpdump
- Uses a rule-based language to match specific traffic patterns

Three operating modes:

- Sniffer mode — prints packets to screen
- Packet Logger mode — logs packets to disk
- Network IDS mode — matches rules and generates alerts

Forensic Relevance

- Snort alerts serve as network-based digital evidence
- Alert logs contain: timestamp, source/destination IP and port, protocol, rule ID, and rule description
- Snort PCAP logs can be opened directly in Wireshark
- Used as the first alert mechanism in Case Study 2 (corporate piracy detection)
- Snort rules can be tuned to detect specific attack signatures relevant to the investigation

Summary: Network Forensics Workflow

1	2	3	4	5
Identify	Plan	Acquire	Analyse	Report
Receive incident report. Define scope and objectives. Identify stakeholders.	Apply OSCAR. Identify evidence sources. Obtain authorisation. Document the plan.	Passive/active capture. Physical tapping. Log collection. Document every action taken.	BPF/Wireshark filtering. Log correlation. Timeline construction. Data reconstruction.	Factual, defensible report. Chain of custody documentation. Recommendations.

Key Terminology Revision

Term	Definition
Digital Evidence	Any information stored or transmitted in binary form that may be relied upon in court
Network-Based Digital Evidence	Evidence produced as a result of communications over a network
Chain of Custody	Documentation of every person who has touched evidence and every action taken on it
OSCAR	Obtain, Strategize, Collect, Analyze, Report — the network forensic investigation methodology
Passive Acquisition	Evidence gathering without emitting data at Layer 2 and above
Active Acquisition	Evidence collection by interacting with live network devices — modifies the environment
Inline TAP	Hardware device that copies network traffic at a point without disrupting normal flow
Vampire Tap	Connector that bites into 10BASE5 coaxial cable to

Key Terminology Revision (continued)

Term	Definition
PCAP	Standard file format for storing captured network packets — universally supported by analysis tools
BPF	Berkeley Packet Filter — powerful language for filtering packets at Layers 2 through 4
Snaplen	The number of bytes of each captured frame that tcpdump records; use 0 for full capture
SIEM	Security Information and Event Management — centralised log aggregation and analysis platform
NetFlow	Cisco protocol that records flow metadata (IPs, ports, bytes) without capturing payload content
IPFIX	IP Flow Information Export — IETF standardised successor to NetFlow
File Carving	Extracting files from raw data or packet captures based on file signatures and structures
Admissibility	The legal standard determining whether evidence can be

Reference Books

Prescribed References for Unit 5 (NFSU Syllabus)

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Unit 5: Network Forensics

Questions and Discussion

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